बहु-विषयक क्षेत्रीय रेल प्रशिक्षण संस्थान ,
धनबाद (भूली)

MULTI-DISCIPLINARY ZONAL RAILWAY
TRAINING INSTITUTE, DHANBAD (Bhuli)

PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

(As per USR – 9.12/4)

PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

In the event of failure of all automatic signals occurring in an area consisting of two of more stations worked under Automatic Block System in single line and when line clear cannot be taken by any of the following means, viz.

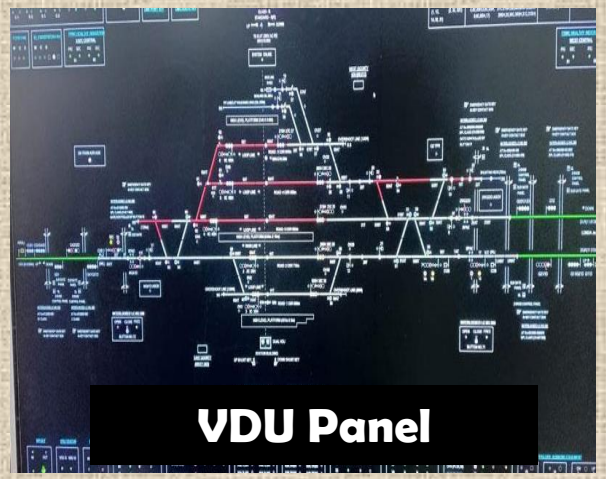
1. Track circuit and Axle counter where indication of the section between two stations is available on VDU panel.



TRACK CIRCUIT



AXLE COUNTER



VDU Panel

PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

2



Station to Station fixed telephones

3



Railway auto phone & BSNL/MTNL phone with caller ID.

4



Railway control telephone

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5



VHF set under special instructions, but not as the sole means of communication on sections where passenger trains run.

6



Fixed Telephone with CUG sim which have been provided to the station only for official use by the on-duty Station Master.

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In the above circumstances trains work in the following manner:

- **The SM who has train to despatch through the effected section, first of all he has to open communication by sending any of the following sources available in order of preference:**

LIGHT ENGINE



TRAIN ENGINE



MOTOR TROLLEY



RAIL MOTOR CAR



TOWER WAGON



EMPTY EMU RAKE



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The Station Master on duty shall also satisfy himself that the Loco Pilot/Motorman/Train Manager/Station Master being sent to open communications thoroughly understands the rules for working of trains during total failure of communications on the single line. If they are not conversant with the Rules for working of trains during total failure of communications on single line, the Station Master on duty shall explain these rules to such staff.



PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

THE LOCO PILOT WILL BE GIVEN FOLLOWING FORMS : -

PHASE- I

WHILE GOING TO OPEN COMMUNICATION OR WHILE GOING TO BRING LINE CLEAR FOR AN AWAITING TRAIN:

i. **T/B 602 - It includes following items :**

   a) **An 'Authority to Proceed Without Line Clear'.**

   b) **A Caution Order,**

      i. **15 KMPH – During day and view ahead is clear.**

      ii. **10 KMPH – During night or when view ahead is not clear.**

      iii. **WP – During thick foggy weather, the train will be piloted by two men with red hand signal and detonator.**

PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

c) A Line clear enquiry message for the waiting train.

d) A Conditional Line Clear Message to the Station Master of the block station at the other end of the affected section permitting him –

i. To return the Light Engine/Train Engine, either light or attached to a train waiting to be despatched from his station, or attached with another engine; or

ii. To return Tower Wagon/Diesel Car/Rail Motor Car/EMU Rake running by itself; or

iii. To return Motor Trolley/ Trolley/ Cycle Trolley/ Moped Trolley either running by itself or loaded in a train waiting to be despatched from his station.

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ii. **T/A -912- - It includes :**

- **Authority to pass automatic/semi automatic/manually operated/gate stop signals.**
- **Number of all signals to be thus passed.**

- **Acknowledgement by Loco Pilot/ Motorman/ Train Manager/ Station Master going to open communications - signing on original and carbon copy of T/B – 602 and T/A-912 in token of his having understood its contents.**

T/B - 602

UP/ DOWN **EASTERN RAILWAY** Form No. T/ B 602

Sr.No_____

AUTHORITY FOR OPENING COMMUNICATION DURING TOTAL INTERRUPTION OF COMMUNICATION ON SINGLE LINE SECTION

(Loco Pilot / Train manager / Station Master's Record)

Station_____ Time:_____hours____mins
Date:_____
To
The Loco Pilot/Motorman of Engine/ Self propelled Vehicle_____

AUTHORITY TO PROCEED WITHOUT LINE CLEAR

1. You are hereby authorised to proceed cautiously without Line clear from station_____to station _____with your Engine / Self-propelled Vehicle.
2. You shall bring your Engine / Vehicle to a stand outside the first stop signal of the next block station thereafter be guided by the instructions from the Station Master of that Station.
3. Train No._____was the Last Train over the section. It left/ arrived* here at_____hours_____ minutes.
4. This authority is given due to:
   (a) Total interruption of communication with Station_____from_____hours_____min of Date_____
   (b) _____

* AUTHORITY TO PASS SIGNALS IN 'ON' POSITION

You are authorised to pass the Signal/s No.& description_____in "ON" position, **speed NOT exceeding 15 kmph** observing hand signals at the foot of the signal post/s, if it protects points.

Strike out for Automatic Block Signaling

T/B - 602

CAUTION ORDER

1. You are permitted to run your Engine/ Self propelled Vehicle between Station_____To _____station_____with **the speed of 15 kmph in day and when view ahead is clear and 10 kmph in night or when view ahead is not clear.**
2. You are expressly warned to observe every caution while approaching level crossing gate/ s situated between block section and whistle frequently.
3. In addition following caution orders are in force in block section:

Sr No	Station between		Kilometer		Speed kmph	Cause/ Remarks
	From	To	From	To		
1.						
2.						
3.						
4.						

LINE CLEAR ENQUIRY MESSAGE

To
The Station Master_____Message No____
On return of_____will line be clear and kept clear for train nowaiting to proceed?

CONDITIONAL LINE CLEAR MESSAGE

To
The Station Master_____Message No
On arrival of_____train at your station, line will be clear and kept clear for Train Engine to returnwith / without attached to a Train or another Engine or Self Propelled Vehicle / Trolley etc (Complete particulars.)
Private Number (in words)_____(in figures)_____
I have understood the contents of this. _____

_____ _____
Signature of Loco Pilot / Motorman Signature of Train Manager Signature & Stamp of Station Master

Time: Hours min

Instructions: 1. Strike out whichever is not applicable.
2. Loco Pilot proceeding on this Authority must observe instructions mentioned at the back of this form.

T/B - 602

IMPORTANT INSTRUCTIONS

Motorman / Loco Pilot / Train Manager proceeding on this authority must observe the following precautions

1. Speed must not exceed 15 kmph over the straight line with clear view and 10 kmph when approaching or passing any portion of the line where the view ahead is not clear due to curve, obstruction, rain, fog or any other cause subject to the observance of speed restriction imposed and speed over facing points restricted to 15 kmph.

2. Both by day and by night, tunnel must not be entered until the Loco Pilot has ascertained that it is clear. Should there be any doubt on this point, the train should be piloted by a Railway Employee equipped with lighted hand signals and detonators.

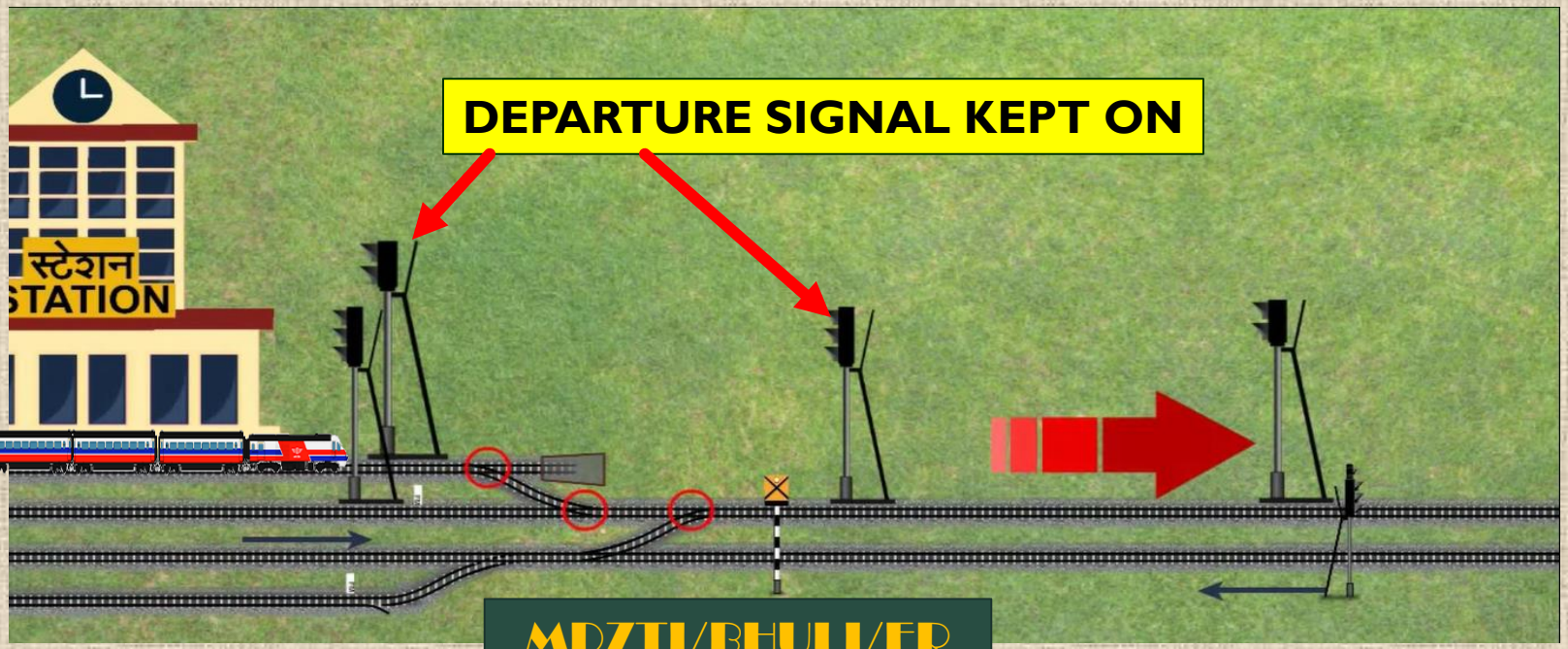
3. A sharp look out must be kept at all times and Loco Pilot must be prepared to stop clear and short of any obstruction which may exist or crop up on the line. Engine whistle must be freely used.

4. When approaching the nominated station ahead, the Loco Pilot must bring his train engine to stand outside the Semi–Automatic / Manual Reception Stop Signal pertaining to right line and outside the Semi–Automatic / Manual Despatch Stop Signal pertaining to wrong line and sound one long whistle and act to the instructions of the Station Master.

5. The form should be handed over to the Station Master of the nominated station at the end of the authorized journey for record.

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- The Semi-Automatic/Manual Despatch Stop Signal shall not be taken 'OFF', while permitting an engine or self-propelled vehicle or other vehicle to proceed to the next station on 'Authority for opening communication during total interruption of communication on single line section'.



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- After the light engine/ vehicle etc have dispatched, no other train/engine/self propelled vehicle/ other vehicle shall be allowed to proceed in the same direction and no obstruction shall be allowed beyond the outermost facing point until the light engine/ vehicle returns.



PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

DUTIES OF CREW WHILE ON RUN TOWARDS OTHER END STATION: -

a. Switch ON Flasher light



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DUTIES OF CREW WHILE ON RUN TOWARDS OTHER END STATION: -

b) LP shall proceed at a speed

i. 15 KMPH during day when the view ahead is clear



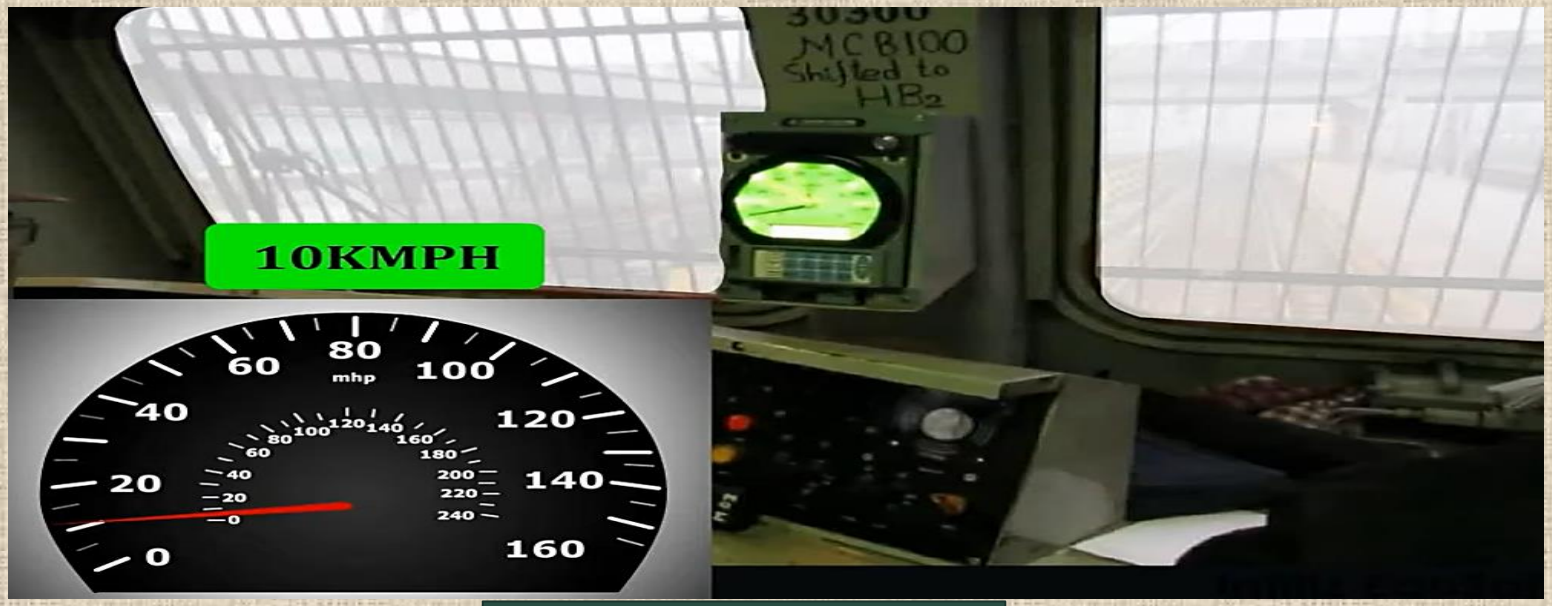
PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

DUTIES OF CREW WHILE ON RUN TOWARDS OTHER END STATION: -

b) **LP shall proceed at a speed**

ii. **10 kilometers per hour during night or when the view is obstructed.**

iii. **Frequent use of whistle/horn.**



PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

DUTIES OF CREW WHILE ON RUN TOWARDS OTHER END STATION: -

b) **LP shall proceed at a speed**

iii. **The speed over facing points, if any, being restricted to 15 KMPH.**



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DUTIES OF CREW WHILE ON RUN TOWARDS OTHER END STATION: -

b) A tunnel must not be entered until its clearance has been ensured both in day and night. If its clearance is doubtful, the engine/vehicles shall be piloted by railway employee equipped with hand signal and detonators. While entering into the tunnel, the headlights, sidelights and tail lights shall also be switched on.



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c) If an engine/vehicle meets in the block section coming from the opposite direction, the Driver/ Motorman/ Guard/ SM shall decide as to which station. They should proceed taking into consideration the following facts: -



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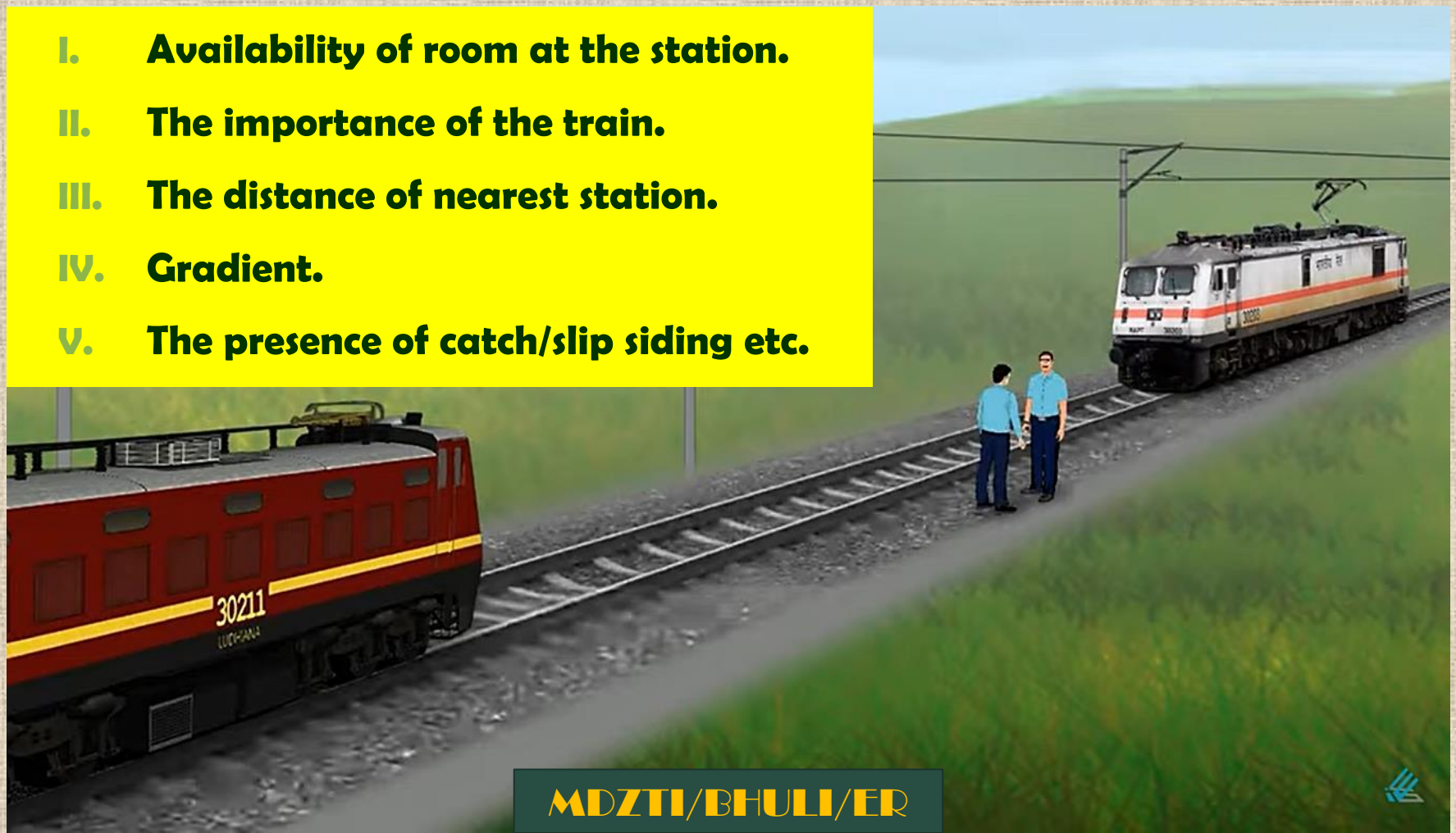
I. Availability of room at the station.

II. The importance of the train.

III. The distance of nearest station.

IV. Gradient.

V. The presence of catch/slip siding etc.



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d) **Before proceeding, the engine or self propelled vehicle shall be coupled (if possible) otherwise they should proceed maintaining an adequate distance between them. In case of push trolley/ motor trolley/ cycle trolley/ moped trolley. It shall be loaded into engine/ diesel car/ rail motor car/ emu rake (if possible)**



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RECEPTION OF TRAIN : -

i. On reaching near the next station engine/ vehicle shall stop at the FSS and should sound long whistle to apprise the station staff who will arrange to receive it by taking 'off' signal/ signals or otherwise.



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RECEPTION OF TRAIN : -

ii. After the train has been received at the station T/B 602 will be handed over to SM on duty.

iii. On receiving the above mentioned paper the SM will arrange to return the engine/ vehicle with or without train (if any) as instructed in form.



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PHASE II

RETURNING OF LIGHT ENGINE WITH LINE CLEAR (WITH OR WITHOUT LOAD) :

THE SM WILL ISSUE THE FOLLWING FORMS: -

1) T/G 602 (for UP) or T/H 602 (for DN) – CLCT
2) T/409 – Caution order, OR T/A 409 – Nil caution order
3) T/E 602 - Line clear enquiry message (if required)
4) T/F 602 – Conditional line clear reply message.(by TM/LP)
5) T/A-912 – Authority to pass automatic/semi automatic/manually operated/gate stop signals.

- *Note : - Authority to pass signal at ON in T/B - 602 should be Stroked out for Automatic Block Signaling.*

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- The SM will hand over the mentioned form to the Driver and dispatch the Engine/Train.
- Train will run as per sectional speed.
- On reaching the next Station the LP/TM of the Train / Light Engine will hand over the following forms to SM:

  a) T/G 602(UP) or T/H 602(DN)

  b) T/E 602 (if given)

  c) T/F 602

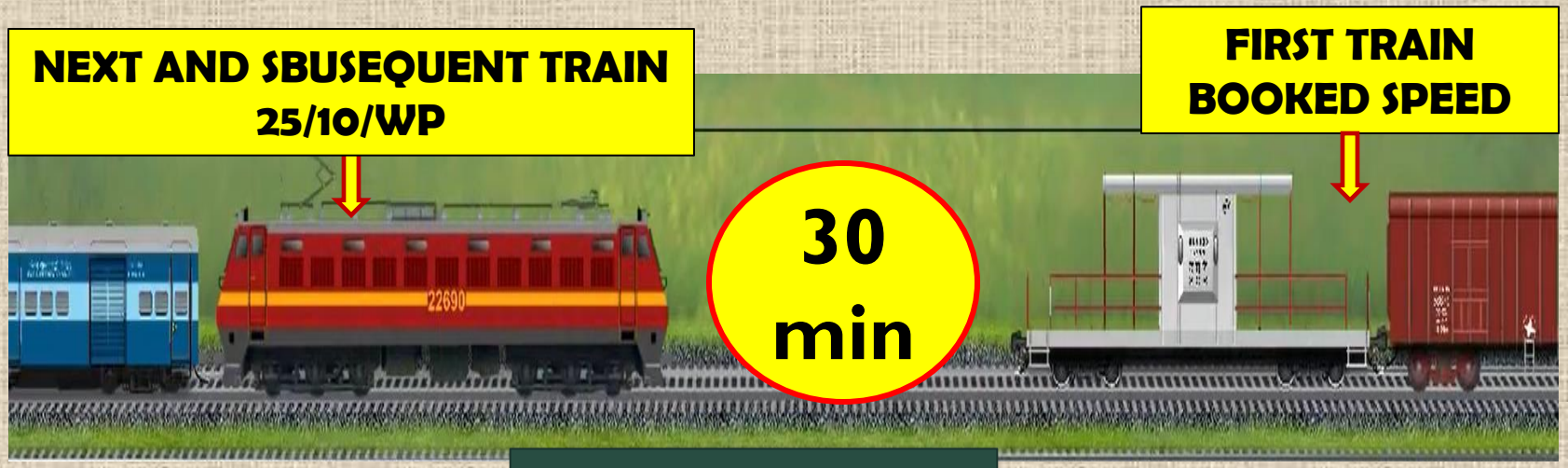
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IIIrd Phase

- On getting the relevant paper, the SM will arrange to dispatch the train (or trains) for which line clear is received, by issuing the following papers:-
  a) T/G 602(up) or T/H 602(DN)
  b) T/A- 912: to pass automatic/semi automatic/manually operated/gate stop signals.
  c) T/409 or T/A 409
  d) T/E 602 to Train Manager (if required)
  e) T/F 602 (if required)

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- After handling over the above mentioned form the first train shall finally be despatched with sectional speed next and subsequent train with speed restriction of 25 kmph (straight line & visibility clear)/10 (curve or visibility not clear) after interval of 30 minutes.



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NOTE:

1. In case train is disabled in block section it should be protected by placing detonators at 250m, 500m & 510 m in rear by Guard.



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NOTE:

2. Normally, backing of train is prohibited but In case of emergency it should be protected by placing detonators at 250m, 500m & 510 m in rear by Guard first then backing should be done.



PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

- Trains will continue to work on this system until any one of the means of communications mentioned earlier is restored by the Competent Authority and when anyone of the means of communication is established the form T/I 602 is duly filled up.



PROLONG FAILURE OF SIGNALS & COMMUNICATION NOT AVAILABLE ON SINGLE LINE IN AUTOMATIC BLOCK SYSTEM

- **PRECAUTION BEFORE RESTORATION:**

i. Complete arrival of last train ensured.

ii. Information to section controller.

- **NOTE:** If communication with Section Controller has not got restored along with restoration of communication between two stations, the Section Controller shall be advised of the position immediately on restoration of communication with him.

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सेवार्थ जाइये